

JEE MAIN REVISION MASTER: ELECTROMAGNETIC WAVES

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 166 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Electromagnetic Waves - Core Application	166 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Electromagnetic Waves - Core Application (Very High Yield — 166 PYQs)

Core Principle: Key problem pattern in Electromagnetic Waves.

■ ■ Caution / JEE Pitfall: Watch for algebraic and sign errors.

■ Speed Trick: Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Morning Shift] Q#1 (MCQ): "A point light source emits E.M. waves in free space. A detector, placed at a distance of L_m , measures the intensity as I_0 . The dete..." [Ans: B]
- [2026 Evening Shift] Q#1 (MCQ): "An electromagnetic wave travels in free space along the x-direction. At a particular point in space and time, $\vec{B}=2 \times 10^{-7}\hat{y}$..." [Ans: N/A]
- [2026 Morning Shift] Q#1 (MCQ): "text { Match the LIST-I with LIST-II } List-I List-II A. Radio-wave I. is produced by Magnetron valve B. Micro-wave II. due to change in th..." [Ans: C]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 6th April Morning Shift (Q#1)

A point light source emits E.M. waves in free space. A detector, placed at a distance of L m, measures the intensity as I_0 . The detector is now shifted to another location on the same spherical surface ensuring the angle between original location and new location as 45° . The measured intensity at new location will be _____.

- (A) $\frac{I_0}{4}$
- (B) I_0 **[Correct]**
- (C) $\frac{I_0}{\sqrt{2}}$
- (D) $\frac{I_0}{2}$

Correct Answer: B

Solution Analysis: The intensity (I) of electromagnetic waves from a point source in free space follows the inverse square law. $I = \frac{P}{4\pi r^2}$ Where P is the power of the source and r is the distance from the source. Initial distance is L and the intensity recorded is I_0 . It is given that the detector is shifted to another location on the same spherical surface. On a spherical surface all points are at a constant distance r from the center (the source). Since the detector remains on the same spherical surface, its distance from the point source remains L . The angle (45°) represents